Task 01 – Welcome Banner

Question: Write a program to print the welcome banner, numbers, expressions and total fee.

Code

```
#include <iostream>
using namespace std;
int main()
{
   cout << "Hello World!" << endl;
   cout << "University of Central Punjab\nFaculty of Information
Technology\nOrientation Day 2026\n";
   cout << "Welcome!" << endl;
   cout << 2026 << endl;
   cout << 45 << endl;
   cout << 45 * 3 << endl;
   cout << 1000 + 500 - 250 << endl;
   cout << "Total fee: " << 45000 + 9000 << endl;
   return 0;
}
```

Sample Output

```
Hello World!
University of Central Punjab
Faculty of Information Technology
Orientation Day 2026
Welcome!
2026
45
135
1250
Total fee: 54000
```

Task 02 – Debug the Code

Question: Correct the syntax errors and calculate the average of 60, 70 and 80.

Code

```
#include <iostream>
using namespace std;
int main()
{
   cout << "Orientation Day" << endl;
   cout << "Total students: " << 450 << endl;
   cout << "Seats left: " << 500 - 450 << endl;
   cout << "Have a nice day" << endl;
   cout << "Average: " << (60 + 70 + 80) / 3 << endl;
   return 0;
}
```

Sample Output

Orientation Day
Total students: 450
Seats left: 50
Have a nice day
Average: 70

Task 03 – Expression Quiz

Question: Print the results of the six arithmetic expressions and use brackets to make expression 6 equal 8.

Code

```
#include <iostream>
using namespace std;
int main()
{
   cout << "Expression 1: " << 10 + 5 * 2 << endl;
   cout << "Expression 2: " << (10 + 5) * 2 << endl;
   cout << "Expression 3: " << 20 - 5 - 3 << endl;
   cout << "Expression 4: " << 100 / 7 << endl;
   cout << "Expression 5: " << 100 % 7 << endl;
   cout << "Expression 6: " << 2 * 3 + 8 / 4 - 1 << endl;
   cout << "Expression 6 with brackets: " << 2 * (3 + 8 / 4 - 1) << endl;
   return 0;
}
```

Sample Output

```
Expression 1: 20
Expression 2: 30
Expression 3: 12
Expression 4: 14
Expression 5: 2
Expression 6: 7
Expression 6 with brackets: 8
```

Task 04 – Admission Form

Question: Declare the given student variables, display them, update semester and fee paid, and calculate remaining fee and percentage paid.

Code

```
#include <iostream>
using namespace std;
int main()
{
   int rollNo = 1045, semester = 1;
   char section = 'B';
   double cgpa = 3.5;
   int totalFee = 45000, feePaid = 20000;

   cout << "Roll No: " << rollNo << endl;
   cout << "Semester: " << semester << endl;
   cout << "Section: " << section << endl;
   cout << "CGPA: " << cgpa << endl;
   cout << "Total Fee: " << totalFee << endl;
   cout << "Fee Paid: " << feePaid << endl;

   semester = semester + 1;
   feePaid = feePaid + 5000;
   int remainingFee = totalFee - feePaid;
   int percentagePaid = feePaid * 100 / totalFee;

   cout << "Semester: " << semester << endl;
   cout << "Fee Paid: " << feePaid << endl;
   cout << "Remaining Fee: " << remainingFee << endl;
   cout << "Percentage Paid: " << percentagePaid << endl;
   return 0;
}
```

Sample Output

```
Roll No: 1045
Semester: 1
Section: B
CGPA: 3.5
Total Fee: 45000
```

Fee Paid: 20000
Semester: 2
Fee Paid: 25000
Remaining Fee: 20000
Percentage Paid: 55

Task 05 – Fee Calculator

Question: Take age, credit hours and fee per credit hour as input. Calculate total fee, discounted fee, birth year and the year the student turns 25.

Code

```
#include <iostream>
using namespace std;
int main()
{
   int age, creditHours, feePerCreditHour;
   cout << "Enter your age: ";
   cin >> age;
   cout << "Enter credit hours: ";
   cin >> creditHours;
   cout << "Enter fee per credit hour: ";
   cin >> feePerCreditHour;

   int totalFee = creditHours * feePerCreditHour;
   int discountedFee = totalFee - 2000;
   int birthYear = 2026 - age;
   int yearTurning25 = 2026 + (25 - age);

   cout << "Total fee: Rs " << totalFee << endl;
   cout << "Fee after discount: Rs " << discountedFee << endl;
   cout << "Year of birth: " << birthYear << endl;
   cout << "Year you turn 25: " << yearTurning25 << endl;
   return 0;
}
```

Sample Output

```
Enter your age: 19
Enter credit hours: 15
Enter fee per credit hour: 8000
Total fee: Rs 120000
Fee after discount: Rs 118000
Year of birth: 2007
Year you turn 25: 2032
```

Task 06 – Bookshop Bill

Question: Write a program from the flowchart. Use price 250 and quantity 3 for the sample output.

Code

```
#include <iostream>
using namespace std;
int main()
{
   int price, quantity;
   cout << "Enter book price: ";
   cin >> price;
   cout << "Enter quantity: ";
   cin >> quantity;

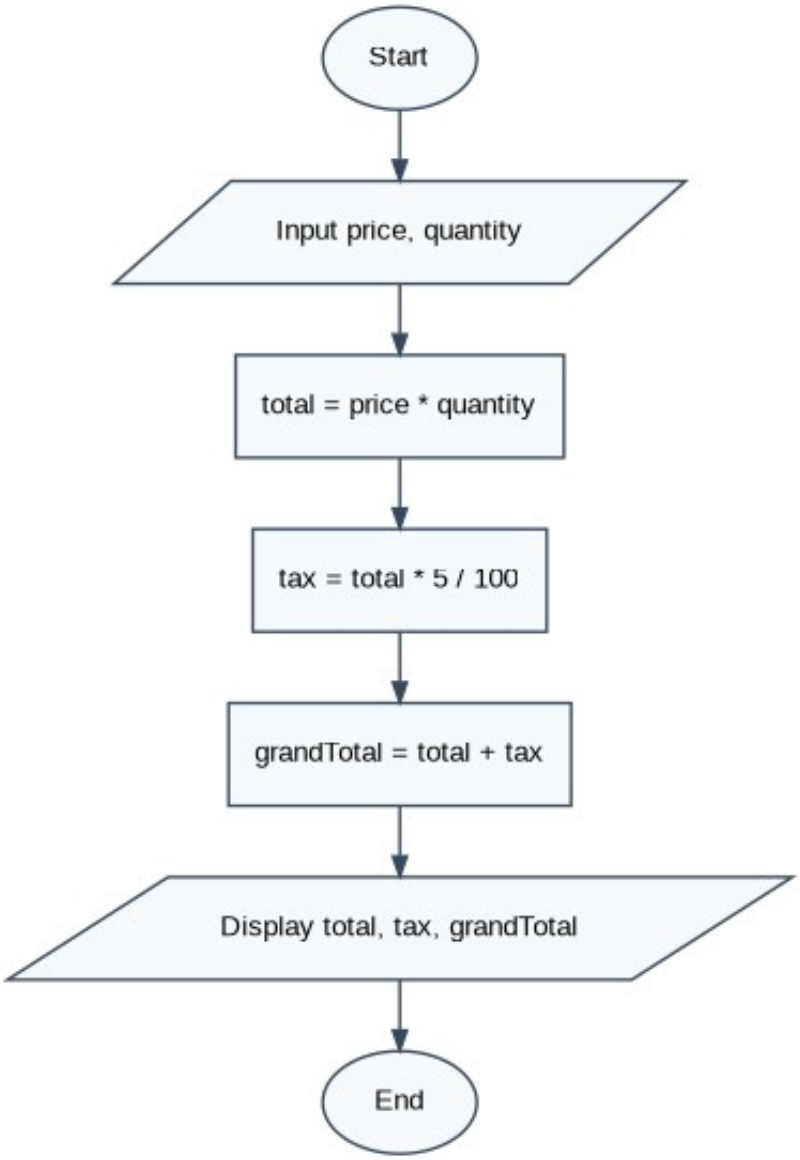
   int total = price * quantity;
   int tax = total * 5 / 100;
   int grandTotal = total + tax;

   cout << "Total: " << total << endl;
   cout << "Tax: " << tax << endl;
   cout << "Grand total: " << grandTotal << endl;
   return 0;
}
```

Sample Output

```
Enter book price: 250
Enter quantity: 3
Total: 750
Tax: 37
Grand total: 787
```

Flowchart


Start
Input price, quantity
total = price * quantity
tax = total * 5 / 100
grandTotal = total + tax
Display total, tax, grandTotal
End

Task 07 – Weather App

Question: Take temperature in Celsius, convert it to Fahrenheit and print both values.

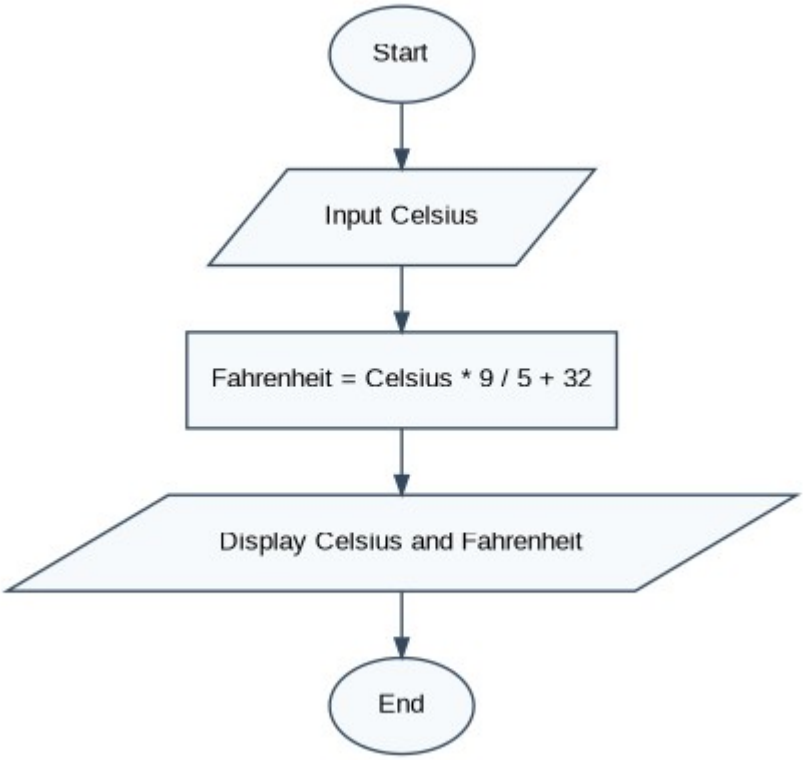
Code

```
#include <iostream>
using namespace std;
int main()
{
   int celsius;
   cout << "Enter temperature in Celsius: ";
   cin >> celsius;
   int fahrenheit = celsius * 9 / 5 + 32;
   cout << "Temperature in Celsius: " << celsius << endl;
   cout << "Temperature in Fahrenheit: " << fahrenheit << endl;
   return 0;
}
```

Sample Output

Enter temperature in Celsius: 100
Temperature in Celsius: 100
Temperature in Fahrenheit: 212

Flowchart


Start
Input Celsius
Fahrenheit = Celsius * 9 / 5 + 32
Display Celsius and Fahrenheit
End

Task 08 – Flowchart with a Decision

Question: Draw a voting eligibility flowchart. Also identify the ticket for ages 8, 12 and 30 in the cinema flowchart.

Sample Output

Voting flowchart: Start → Input age → Is age >= 18?
Yes → Output "Eligible to vote"
No → Output "Not eligible to vote"
Both paths → End

Cinema answers:
Age 8: Half ticket
Age 12: Full ticket
Age 30: Full ticket

Flowchart

